concerning Presentation at GE KAPL

Oct 2 Thurs. FR-7-8521 F. Fisher
P. Hilgard

2:00 or 2:30

426 Franklin St.

come for lunch 12:00 2 hrs, by taxpike.

KAPL 2 meetings

Logic group — Naval Physicists — gen. discussion of problems which can be shown other CA better than other computers

Analog group — mechanic function

Dick Ekblad - Mgr adv. Development Activities
Charles Kerber - " 704 Opns
Harry Hunter - Analyst - Pres. & future Computer Reqs.
Don McMillan - Mgr 704 Programming
Geo. Habettler - Math. consultant

Harry Smith - AEC Representative

Notes on Reactor Calc.

ϕ_{m-1} write

C_m ϕ_m compute ϕ_m'

C_{m+1} read ϕ_{m+1}

ϕ_{m+2} read

(24 × 4 × 32) 8

= 32K

bring more (by?)

ϕ_{n+1}' W

C_n $\phi_n \rightarrow \phi_n'$

R read C_{n+1} ϕ_{n+1}

R ϕ_{n+2}

2500 word block

ϕ_{n-1} ϕ_{n-1}' ϕ_{n-1}'' W

C_n ϕ_n ϕ_n'

C_{n+1} ϕ_{n+1}

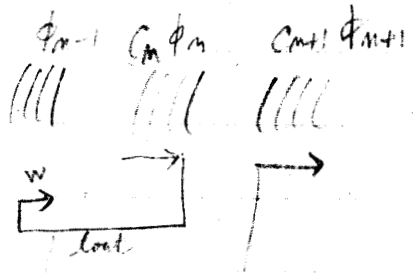
R' C_{n+1} ϕ_{n+2}

R'

calc. 2 cycles

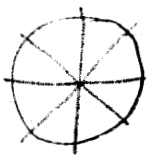
Time per cycle		(+12) compute	I/O	Total
8μs small mem	(1)	35. μs	64. μs	64
8μs large mem 2	(2)	35.	32.	35
4μs 1	(3)	35.	36	36
2	(4)	35.	18	35

Tracks on disk 4096



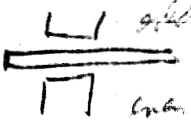
1024 on track

sectors $\frac{1}{8}$ rotation on foot



1½ tracks (not rolled)

32 + 32 + 16 = 80
1 -- 32 + 16 = 48
0 + 32 + 16 = 48



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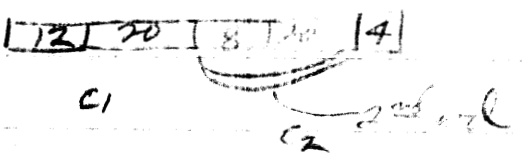
pack



3 Cyl
1 S
1 4
5 seeks

F LD
F.MPY
F ADD
FSTO
VF L Load
F.MPY
F ADD
F STO

$\frac{90K}{5}$ = 18,000 pts per group in mem, adjacent

20×30×30 mesh

0.360000 sec to write old group
0.36 to read new
0.72 sec I/O

(18,000 pts)(50 μsec) = 0,900 000 sec per iteration
counting scale etc.

16 planes

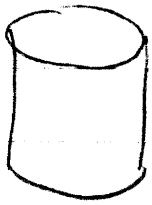
Block

Example: consider 4 groups,

$8 \overline{)256\text{K}}$ $\doteq$ (16 × 45 × 45 mesh) 4 groups. (3 mesh)

32K in mem.

→ can just flush C's thru & keep 4's in.
(4 S's)



6 C's = 194,400

32,400. plane 4's in mem
32,400. S's in mem
64,800 ... of mem. needed

other 3 groups (194,400 + 32,400 = 259,200.)

3(259,200) = 777,600.

from other 4 32,400.

1,036,800

(16 planes)(2025)

S ϕ C C'

32K 32K 12K 12K (48 μs

4K words per block

3 blocks every 8,2 μs (μs),

C ϕ ϕ'

C ϕ ϕ

C ϕ